

Kan

Kan extension motivation

The *Kan extension* of a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ with respect to a functor

$$\begin{array}{c} \mathcal{C} \\ \downarrow p \\ \mathcal{C}' \end{array}$$

is, if it exists, a kind of *best approximation* to the problem of finding a functor $\mathcal{C}' \rightarrow \mathcal{D}$ such that

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ p \downarrow & \nearrow & \\ \mathcal{C}' & & \end{array}$$

hence to *extending* the *domain* of F through p from $\mathcal{C}$ to $\mathcal{C}'$.

Kan lift motivation

Similarly, a *Kan lift* is the best approximation to lifting a morphism $F: \mathcal{C} \rightarrow \mathcal{D}$ through a morphism

$$\begin{array}{c} \mathcal{D}' \\ \downarrow \\ \mathcal{D} \end{array}$$

to a morphism $\hat{F}$

$$\begin{array}{ccc} & & \mathcal{D}' \\ & \nearrow \hat{F} & \downarrow \\ \mathcal{C} & \xrightarrow{F} & \mathcal{D}. \end{array}$$

[2]

If you think of functor composition as a form of multiplication, Kan extensions are an attempt to construct inverses of this multiplication. But unlike multiplication, composition is not symmetric, so we have *extensions* that attempt to undo precomposition, and *lifts* that do the same for postcomposition. Furthermore, there rarely is a single inverse to any form of composition, so we have the parsimonious *right* extensions and lifts, and the generous *left* extensions and lifts. We end up with four combinations that correspond to four different adjunctions:

$$\begin{aligned}(- \circ j) \dashv \text{Ran}_j - \\ \text{Lan}_j - \dashv (- \circ j) \\ (j \circ -) \dashv \text{Rift}_j - \\ \text{Lift}_j - \dashv (j \circ -)\end{aligned}$$

We'll concentrate on the extensions, since we can provide explicit point-wise formulas for them in cases that are of interest to us, that is in **Cat** and in **Prof**.

不要把

$$\text{Lan}_j$$

或者

$$\text{Ran}_j$$

理解成真正的 inverse。

通常：

$$\text{Lan}_j(G) \circ j \neq G$$

甚至：

$$\text{Ran}_j(G) \circ j \neq G.$$

它们只是分别通过：

$$G \Rightarrow (\text{Lan}_j G) \circ j$$

和

$$(\text{Ran}_j G) \circ j \Rightarrow G$$

给出一种 **universal approximation**。

所以作者所谓：

generous vs. parsimonious

真正应该理解成：

Left Kan extension initial universal solution 自由、尽可能扩张 ↓ generous	Right Kan extension terminal universal solution 保守、尽可能约束 ↓ parsimonious
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而这个直觉又恰好与：

$\text{Lan} \sim \text{colimit}$	$\text{Ran} \sim \text{limit}$
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相吻合。

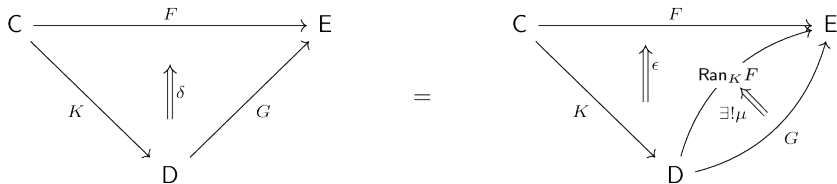
left Kan extension

A left Kan extension of a functor $F: \mathcal{C} \rightarrow \mathcal{E}$ along $K: \mathcal{C} \rightarrow \mathcal{D}$ consists of a functor $\text{Lan}_K F: \mathcal{D} \rightarrow \mathcal{E}$ and a unit natural transformation $\eta: F \Rightarrow \text{Lan}_K F \circ K$ such that for any other extension pair $(G: \mathcal{D} \rightarrow \mathcal{E}, \gamma: F \Rightarrow G \circ K)$, there is a unique $\mu: \text{Lan}_K F \Rightarrow G$ yielding the factorization $\gamma = \mu K \circ \eta$. This can be visualized by the pasting diagram



right Kan Extension

What's left is what's on the right, and indeed there is a dual notion of a right Kan extension, consisting of a functor $\text{Ran}_K F: \mathcal{D} \rightarrow \mathcal{E}$ and counit natural transformation $\varepsilon: \text{Ran}_K F \circ K \Rightarrow F$ such that for any other extension pair $(G: \mathcal{D} \rightarrow \mathcal{E}, \delta: G \circ K \Rightarrow F)$, there is a unique $\mu: G \Rightarrow \text{Ran}_K F$ yielding the factorization $\delta = \varepsilon \circ \mu K$.



Building Kan Extensions

We have already discussed the notion that Kan extensions are a kind-of approximation for a true extension, modulo a natural transformation. In particular, a left Kan extension of $F: \mathcal{C} \rightarrow \mathcal{E}$ along $K: \mathcal{C} \rightarrow \mathcal{D}$ is the “best” approximation of an extension of F along K “from the left”, which is to say up to a natural transformation $\eta: F \Rightarrow \text{Lan}_K F \circ K$ initial among all natural transformations $F \Rightarrow GK$ via an arrow $\text{Lan}_K F \Rightarrow G$ “from the left”. Taking this idea of “approximating from the left” to heart, to define the action of $\text{Lan}_K F$ on an object d we may want to consider the action of F on objects in $\mathcal{C}$ that “approximate” d .

In the present context, there are two reasonable ways to determine what objects “approximate” d . One is to choose those objects $c \in \mathcal{C}$ such that $Kc \cong d$. Another is to choose objects $c \in \mathcal{C}$ whose image under K can “get to” d , which is to say there is an arrow $Kc \rightarrow d$. Indeed, considering the idea of approximation most universally, the latter feels more appropriate. Such “approximating” arrows $Kc \rightarrow d$ are exactly the objects of the comma category $K \downarrow d$, which has a canonical projection functor $\Pi^d: K \downarrow d \rightarrow \mathcal{C}$ that takes an arrow $Kc \rightarrow d$ defining an “approximation of d from the left” to the “approximating object” c . We can then map each such object via F , defining a diagram $K \downarrow d \xrightarrow{\Pi^d} \mathcal{C} \xrightarrow{F} \mathcal{E}$ whose image seemingly provides a number of approximations for the image of d “from the left”. How do we collapse these approximations into a single object? Take a colimit!

Theorem. Consider $F: \mathcal{C} \rightarrow \mathcal{E}$ and $K: \mathcal{C} \rightarrow \mathcal{D}$. If $\forall d \in \mathcal{D}$ the colimit

$$\operatorname{colim}(K \downarrow d \xrightarrow{\Pi^d} \mathcal{C} \xrightarrow{F} \mathcal{E}) =: \operatorname{Lan}_K F(d)$$

of the diagram $F\Pi^d$ exists, these data assemble into a functor $\operatorname{Lan}_K F: \mathcal{D} \rightarrow \mathcal{E}$, and the colimit cones embed the unit $\eta: F \Rightarrow \operatorname{Lan}_K F \circ K$, such that $\operatorname{Lan}_K F, \eta$ define the left Kan extension of F along K .

Dually, if $\forall d \in \mathcal{D}$ the limit

$$\operatorname{lim}(d \downarrow K \xrightarrow{\Pi_d} \mathcal{C} \xrightarrow{F} \mathcal{E}) =: \operatorname{Ran}_K F(d)$$

of the diagram $F\Pi_d$ exists, these data assemble into a functor $\operatorname{Ran}_K F: \mathcal{D} \rightarrow \mathcal{E}$, and the limit cones embed the counit $\varepsilon: \operatorname{Ran}_K F \circ K \Rightarrow F$, such that $\operatorname{Ran}_K F, \varepsilon$ define the right Kan extension of F along K .

Adjunction:

$$F \dashv G$$

可以特殊地编码为:

$$G = \mathrm{Lan}_F 1,$$

或者:

$$F = \mathrm{Ran}_G 1.$$

Limit / colimit:

又是 Kan extension 的另一个特殊情形:

$$\mathrm{colim} F = \mathrm{Lan}_! F$$

$$\mathrm{lim} F = \mathrm{Ran}_! F$$

其中:

$$! : \mathcal{C} \rightarrow \mathbf{1}.$$

参考文献 I

 <https://ncatlab.org/nlab/show/Kan+extension>

 [https://bartoszmilewski.com/2026/06/08/
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